

Zhifei (Andy) Li

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Research Interests

My research is on **machine learning systems**, centered on the co-design of efficient algorithms and systems. I am especially interested in **agentic ML systems**: those built **by** agents, where AI autonomously writes and optimizes low-level kernels and full systems, and those built **for** agents, where the systems stack itself is redesigned for AI as a first-class user. My prior work spans resource scheduling, training and inference infrastructure, and storage-efficient retrieval.

Research Experience

Sky Computing Lab

University of California, Berkeley

RESEARCH INTERN, ADVISED BY PROF. ION STOICA

WORKED WITH PROF. JOSEPH E. GONZALEZ, PROF. MATEI ZAHARIA

July 2025 - December 2025

- **SkyNomad: Multi-Region Spot Instance Scheduling** (accepted to NSDI '27)
 - Designed a multi-region spot instance scheduling system, addressing single-region availability bottlenecks for offline workloads, via Unified Cost Model trading off cross-region availability and pricing vs. migration costs
 - Achieved >50% cost reduction over previous SOTA (NSDI '24 best paper), saved \$1,000+ from a \$2,200 training job vs. AWS SageMaker
 - Led project from research formulation to production, drove methodology design, and built simulation framework
- **LEANN: Storage-Efficient Compound AI Systems** (MLSys '26)
 - Co-designed a two-level search that prunes on-the-fly embedding recomputation, cutting vector index storage in RAG pipelines
 - Achieved 97% storage reduction, <5% latency impact; led open-source implementation to 12,900+ GitHub stars
 - Built the system extending FAISS C++, contributing 70% codebase; drove evaluation efforts
- **AI-driven Systems Research**
 - Investigated systems optimization through evolutionary algorithms and LLM-guided design space exploration
 - Led case study in Barbarians at the Gate paper, which demonstrated 30% improvement over the SOTA
 - Co-developed FrontierCS, an open-ended benchmark of 240 problems; built problem specifications and the evaluation harness

ByteDance Seed

Beijing, China

RESEARCH INTERN, LLM INFERENCE

June 2026 - August 2026

Education

Princeton University

Princeton, NJ, USA

PH.D. IN COMPUTER SCIENCE, ADVISED BY PROF. RAVI NETRAVALI AND PROF. TRI DAO

September 2026 - Present

University of California, Berkeley

Berkeley, CA, USA

EXCHANGE STUDENT, COMPUTER SCIENCE

August 2024 - December 2024

- Took CS294-162 Machine Learning Systems, a graduate seminar

Renmin University of China (Ranked 23rd globally on CSRankings 2025)

Beijing, China

BACHELOR'S IN COMPUTER SCIENCE, **TURING HONORS CLASS**

September 2022 - June 2026

- GPA: 3.8/4.0 (Top 5%)

Publications

SkyNomad: On Using Multi-Region Spot Instances to Minimize AI Batch Job Cost

Zhifei Li*, Tian Xia*, Ziming Mao, Zihan Zhou, Ethan J. Jackson, Jamison Kerney, Zhanghao Wu, Pratik Mishra, Yi Xu, Yifan Qiao, Ion Stoica

NSDI '27; [PDF](#)

LEANN: A Low-Storage Vector Index for Personal Devices

Yichuan Wang, Zhifei Li, Shu Liu, Yongji Wu, et al., Ion Stoica, Sewon Min, Matei Zaharia, Joseph Gonzalez
MLSYS '26 (BEST PAPER AWARD); [\[PDF\]](#)

PixelRAG: Web Screenshots Beat Text for Retrieval-Augmented Generation

Yichuan Wang*, Zhifei Li*, Zirui Wang, Paul Teiletche, Lesheng Jin, Matei Zaharia, Joseph E. Gonzalez, Sewon Min
NEURIPS '26 (IN SUBMISSION); ICML '26 EMMQA WORKSHOP; [\[WEBSITE\]](#)

Barbarians at the Gate: How AI is Upending Systems Research

Audrey Cheng*, Shu Liu*, Melissa Pan*, Zhifei Li, Bowen Wang, Alex Krentzel, et al., Matei Zaharia, Ion Stoica
ARXIV PREPRINT, 2025; [\[PDF\]](#)

FrontierCS: Evolving Challenges for Evolving Intelligence

Qiuyang Mang*, Wenhao Chai*, Zhifei Li*, Huanzhi Mao*, et al., Ion Stoica, Jingbo Shang, Zhuang Liu, Alvin Cheung
ICML '26; [\[PDF\]](#); [\[WEBSITE\]](#)

SkyWalker: A Locality-Aware Cross-Region Load Balancer for LLM Inference

Tian Xia, Ziming Mao, Jamison Kerney, Ethan J. Jackson, Zhifei Li, Jiarong Xing, Scott Shenker, Ion Stoica
EUROSYS 2026; [\[PDF\]](#)

Open-Source Projects

LEANN: the Smallest Vector Index in the World

 [LEANN](#) 12.9k ★

ENJOY 97% STORAGE SAVINGS FOR RAG APPLICATION ON YOUR PERSONAL DEVICE

September 2024 - Present

- Led research-to-production translation of [LEANN](#) from prototype to production-ready open-source Python package with CI/CD pipeline, grew to [12,900+ GitHub stars](#) with 20 active external contributors and 40k+ downloads
- Drove technical outreach including blog posts social media campaign achieving 600k+ views

PixelRAG: Web Screenshots Beat Text for RAG

 [PixelRAG](#) 9.9k ★

SEARCH ANY DOCUMENT BY HOW IT LOOKS, NOT JUST THE TEXT IT CONTAINS

January 2026 - Present

- Co-first-authored and led the project: 18.1% higher accuracy across 6 benchmarks at 10× lower token cost
- Shipped it as developer infrastructure: a hosted search API over a full-Wikipedia index, plus [pixelbrowse](#), a coding-agent skill giving Claude Code / Codex / Cursor visual page access; [9,900+ GitHub stars](#)

SkyPilot: Run AI on Any Infra

 [SkyPilot](#) 10.6k ★

FRAMEWORK FOR RUNNING ML/AI WORKLOADS ACROSS ANY CLOUD INFRASTRUCTURE

September 2024 - Present

- Top 10 contributor; created 70+ issues and merged 50+ pull requests; contributed 30,000+ lines of code changes
- Implemented [High Availability Controller](#) for SkyServe control plane; adopted by startups including Hypermode, IP Copilot and Liner

Services

USENIX ATC '25 Artifact Evaluation Committee

REVIEWER

May 2025

Introduction to Computer Systems (ICS)

HEAD TEACHING ASSISTANT

Fall 2024, Spring 2025

- Led 6 TAs in teaching systems curriculum that covered cache hierarchies and memory optimization to 200+ students

RUC Computer Association

PRESIDENT

July 2024 - July 2025

- Organized 10+ tech talks, 5000+ total attendees, covering topics from Functional Programming to Rust ecosystem
- Led 100+ members across 6 departments, fostered a startup atmosphere and inclusive environment

Cheese Tech

CO-FOUNDER

September 2023 - October 2024

- Co-founded an AI-native platform for student projects; grew to 1,000+ users across 3 partner institutions within the first year
- Built it around an AI teammate that researches, plans, and executes tasks inside the platform, carrying a project from first idea to shipped result (“Agent for Everyone”)

Honors and Awards

<i>Beijing Municipal Outstanding Graduate</i> (top 5% citywide)	June 2026
<i>Wu Yuzhang Scholarship</i> (RUC's highest student honor, <10 recipients/year)	May 2026
<i>Pioneer Medal, RUC</i> (academic innovation leader; 8 awarded of 106)	May 2026
<i>First Place, National System Capability Competition</i> (among 10,000 from 1,200 schools)	November 2025
<i>Elite Collegiate Award, China Computer Federation</i> (<100 recipients nationally)	August 2025
<i>Dean's Scholarship, Gaoling School of AI</i> (15 recipients out of 2,000)	May 2025
<i>National Scholarship</i> (Top 0.2% nationally)	September 2024
<i>First-Class Scholarship for Social Service</i> (48 recipients out of 35,000)	September 2023
<i>First Prize, National Olympiad in Informatics, Beijing</i> (300 winners)	December 2019

Skills

Coding	C++, CUDA, Python, Rust, TypeScript
Tools	PyTorch, SkyPilot, Ray, Kubernetes, verl, NeMo, Nix, Fish, Typst
Languages	English, Chinese (native), French
Hobbies	Texas Hold'em, Badminton, Chess, Cooking